

BANK MARKETING CAMPAIGN ANALYSIS

Data Analytics Final Project – Documentation, EDA, Insights & Power BI Dashboard

1. Project Overview

This project analyzes the Bank Marketing dataset to understand customer characteristics, marketing campaign behavior, and factors associated with term-deposit subscription. Python is used for data loading, cleaning, preprocessing, exploratory data analysis (EDA), and visualization. Power BI is used to build an interactive dashboard for business-focused reporting.

2. Problem Statement

Banks conduct marketing campaigns to encourage customers to subscribe to term deposits. The objective is to analyze demographic, financial, contact, and previous-campaign information to identify patterns associated with subscription and provide data-driven recommendations for improving campaign targeting.

3. Dataset Selection

Dataset: Bank Marketing (bank.csv)

Rows: 45,211

Columns: 17 (16 input features + 1 target)

Target: y (yes = subscribed, no = not subscribed)

CSV separator: semicolon (;).

4. Dataset Features

Feature	Description
Age	Customer age
Job	Occupation
Marital	Marital status
Education	Education level
Default	Credit in default
Balance	Average yearly account balance
Housing	Housing loan
Loan	Personal loan
Contact	Communication method
Day	Day of month of last contact
Month	Month of last contact
Duration	Last contact duration in seconds
Campaign	Number of contacts in current campaign
Pdays	Days since previous campaign contact; -1 means not previously contacted
Previous	Number of previous contacts
Poutcome	Previous campaign outcome
Y	Term-deposit subscription target

TASK 2 – Data Cleaning & Pre-processing

The dataset was validated for structure, data types, missing values, and duplicate records. The dataset contains no missing values and no duplicate rows based on the validation performed. Because the initial objective is EDA, extensive machine-learning preprocessing was not required.

Data Loading and Validation

```
df = pd.read_csv("bank.csv", sep=";")
print(df.shape)
print(df.isnull().sum())
print(df.duplicated().sum())
```

Derived Features

- age_group – groups customers into 18–25, 26–35, 36–45, 46–55 and 56+.
- campaign_category – groups campaign contacts into 1, 2–3, 4–5 and 6+.
- duration_minutes – converts call duration from seconds to minutes.
- subscription – converts y into numeric form: no = 0, yes = 1.

TASK 3 – EDA & Visualizations

The analysis progresses from univariate to bivariate and multivariate analysis. Univariate analysis describes individual variables; bivariate analysis examines relationships between two variables; multivariate analysis examines three or more variables together.

A. Univariate Analysis

- 1. Term Deposit Subscription (y) – shows target distribution and class imbalance.
- 2. Customer Age Distribution (age) – shows the customer age profile.
- 3. Customers by Job (job) – shows occupational composition.
- 4. Education Level Distribution (education) – shows education composition.
- 5. Marital Status Distribution (marital) – shows marital-status composition.
- 6. Account Balance Distribution (balance) – shows distribution, skewness and extreme values.
- 7. Campaign Contacts Distribution (campaign) – shows contact frequency.
- 8. Last Contact Duration Distribution (duration) – shows call-duration patterns.

B. Bivariate Analysis

- 9. Housing Loan vs Subscription – compares subscription by housing-loan status.
- 10. Personal Loan vs Subscription – compares subscription by personal-loan status.
- 11. Subscription Rate by Job – compares subscription rates across occupations.
- 12. Subscription Rate by Education – compares rates across education levels.
- 13. Previous Campaign Outcome vs Subscription – examines the association between previous outcome and current subscription.
- 14. Subscription Rate by Contact Month – examines variation by contact month.
- 15. Age vs Account Balance – examines the relationship between two numerical variables.

C. Multivariate Analysis

- 16. Job + Education + Y – identifies subscription patterns across occupation and education combinations.
- 17. Age+ Balance+ Subscription – compares age groups while considering housing-loan status.

- 18 .Correlation Analysis –age, balance, day, duration, campaign, pdays, previous, subscription

TASK 4 – Documentation, Insights & Presentation

Key Insights

- The dataset contains 45,211 customers and 5,289 subscribed customers, giving an overall subscription rate of approximately 11.7%.
- The target is highly imbalanced, with substantially more customers not subscribing than subscribing.
- Subscription rates vary across occupations and education levels, supporting customer segmentation.
- Previous campaign outcome is useful as a descriptive indicator of current subscription behavior.
- Subscription rates vary across contact months, suggesting that campaign timing can be evaluated.
- Call duration and campaign-contact variables provide useful campaign-performance information, but observed relationships should be interpreted as associations rather than causation.

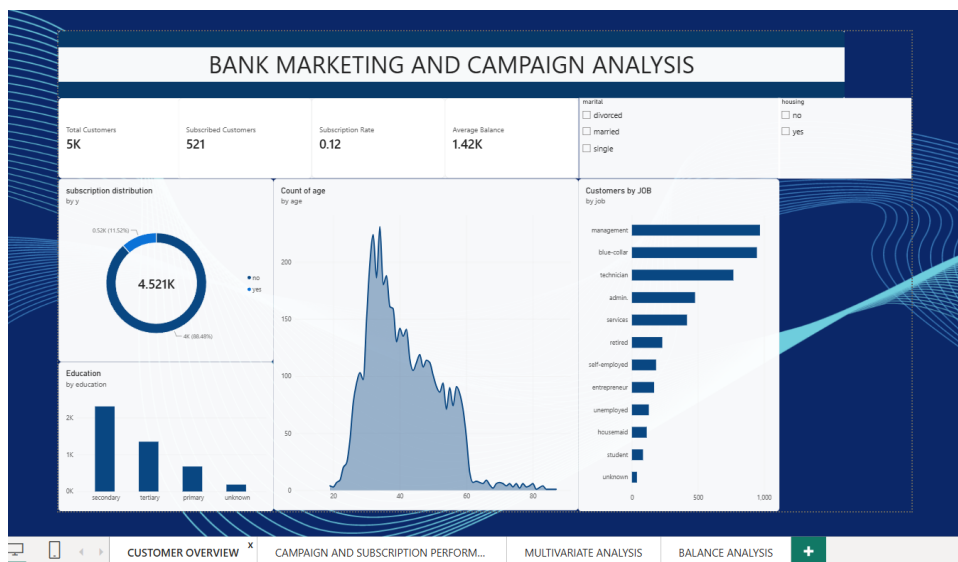
Business Recommendations

- Prioritize customer segments with higher historical subscription rates.
- Use previous campaign outcomes when planning follow-up campaigns.
- Study successful customer conversations to identify effective engagement patterns.
- Review campaign timing and allocate resources toward periods with stronger response rates.
- Monitor repeated contacts and avoid inefficient contact strategies without evidence of engagement.

Power BI Dashboard

The Power BI dashboard presents the analysis through KPI cards, interactive charts, and slicers.

Dashboard 1 - Customer Overview



KPI Cards

- Total Customers – 45,211
- Subscribed Customers – 5,289
- Not Subscribed Customers – 39,922
- Subscription Rate – approximately 11.7%
- Average Balance

Slicers used:

- Job
- Education
- Marital status
- Housing

Charts:

Chart 1 — Subscription Distribution

Visual: Donut chart

Legend → y

Values → Count of y

Chart 2 — Customer Age

Visual: Column chart

X-axis → age_group

Y-axis → Count of age

Chart 3 — Customers by Job

Visual: Bar chart

Y-axis → job

X-axis → Count of job

Chart 4 — Education

Visual: Column chart

X-axis → education

Y-axis → Count of education

Chart 5 — Customers Job by Job

Visual: Column chart

X-axis → Job

Y-axis → Count of Job

This page answers:

Who are the customers?

This page demonstrates univariate analysis.

Dashboard 2- Campaign And Subscription Performance



KPI Cards:

- Total Customers
- Subscribed Customers
- Subscription Rate
- Average Call Duration
- Average Campaign Contacts

Slicers used:

- Month
- Contact
- Poutcome
- Campaign

Charts:

1. Subscription Rate by Job

Visual: Bar chart

Y-axis → job

X-axis → Subscription Rate

2. Subscription Rate by Education

Visual: Column chart

X-axis → education

Y-axis → Subscription Rate

3. Subscription Rate by Month

Visual: Line chart

X-axis → month

Y-axis → Subscription Rate

4. Previous Campaign Outcome

Visual: Bar chart

Y-axis → p out come

X-axis → Subscription Rate

5. Campaign Contacts

Visual: Column chart

X-axis → campaign category

Y-axis → Count of Customers

6. Call Duration

Visual: Column chart

X-axis → duration minutes

Y-axis → Count of Customers

This page answers,

How did the marketing campaign perform, and which factors are associated with subscription?

This page demonstrates bivariate analysis.

Dashboard 3 – MULTIVARIATE ANALYSIS:



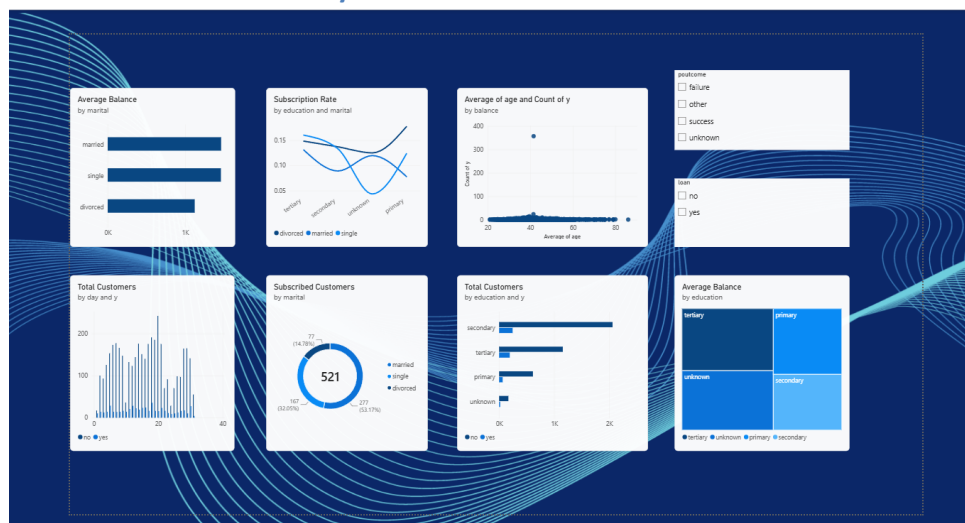
Slicers

- Job
- Education
- Housing
- Loan
- Month
- Poutcome

CHARTS:

- Clustered bar chart: Age by Balance and Y
- Line chart: Previous campaign by month and Previous outcome
- Clustered bar chart: Housing loan by job and personal loan
- Stacked column chart: Personal loan by subscription using loan and Y
- KPI: Subscribed customers and Subscription Rate by Job
- KPI: Total Customers and Subscribed Customers by Campaign

Dashboard 4- Balance Analysis:



Slicers:

- Poutcome
- Loan

Charts:

- Stacked Bar chart – Average Balance by Marital status
- Line chart – Subscription Rate by Education and Marital Status
- Scatter Plot- Age by Balance and Target Variable Y
- Clustered column chart -Total Customers by day and Y
- Donut chart- Subscribed Customers by Marital status
- Clustered bar chart- Total Customer by Education and Y
- Treemap- Average Balance by Education

Core DAX Measures

Total Customers = COUNTROWS (bank)

Subscribed Customers =
CALCULATE (COUNTROWS (bank), bank[y] = "yes")

Subscription Rate =
DIVIDE ([Subscribed Customers], [Total Customers], 0)

Average Balance = AVERAGE (bank[balance])

Average Call Duration = AVERAGE (bank[duration])

Final Conclusion

The Bank Marketing Campaign Analysis transformed raw customer and campaign data into structured analytical insights. Python was used for data validation, preprocessing, feature creation, EDA, and visualization, while Power BI was used for interactive reporting. The analysis shows an overall subscription rate of approximately 11.7% and demonstrates that customer characteristics and campaign-related variables can support segmentation and marketing decisions. The project covers the complete analytics workflow from problem definition and dataset selection through cleaning, EDA, multivariate analysis, dashboard development, insights, and recommendations.

Presentation Story – Short Version

“My project is Bank Marketing Campaign Analysis. I analyzed 45,211 customer records using Python to understand customer characteristics and marketing campaign performance. I performed data validation, created derived features, and conducted univariate, bivariate, and multivariate analysis. The target was term-deposit subscription, with an overall subscription rate of about 11.7%. I then built a Power BI dashboard using KPIs, charts, and slicers to communicate the main patterns. Finally, I identified customer and campaign segments that can help improve marketing targeting and decision-making.”